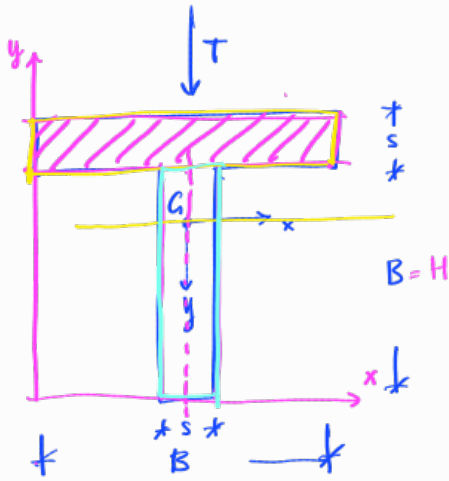


Calcolare la tensione tangenziale massima nell'ala e nell'anima.



① Calcolo del baricentro

$$A = 2Bs$$

$$S_x = B \cdot s \left(B + \frac{s}{2} \right) + Hs \frac{H}{2} = \boxed{B^2 \cdot s} + \frac{Bs^2}{2} + \boxed{\frac{Bs^2}{2}} = \frac{3Bs^2}{2} + \frac{Bs^2}{2}$$

$$y_G = \frac{S_x}{A} = \frac{\frac{3Bs^2}{2} + \frac{Bs^2}{2}}{2Bs} = \frac{\frac{3Bs^2}{2}}{2 \cdot 2Bs} + \frac{Bs^2}{2 \cdot 2Bs} = \frac{3B}{4} + \frac{s}{4} = \frac{1}{4}(3B + s)$$

② Momento di inerzia

$$J_x = \underbrace{\frac{Bs^3}{12} + Hs \left(B + \frac{s}{2} - y_G \right)^2}_{\text{Ala}} + \underbrace{\frac{sH^3}{12} + sH \left(y_G - \frac{H}{2} \right)^2}_{\text{Anima}}$$